

COMPUTING THE PATH TO EMPLOYMENT: UNDERSTANDING THE IMPACT OF COMPUTER KNOWLEDGE IN INDIA AND THE CONTRIBUTING FACTORS

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Abstract

This study examines the influence of computer knowledge on employment in India, considering various socio-economic factors. It explores the correlation between computer literacy and employment probability, the determinants of computer skills, and the differential impact across high-income and low-income states. The analysis utilizes data from the NSS 78th Round and employs logistic and Poisson regression models. Findings suggest a positive relationship between computer knowledge and employment, with education, household expenditure, and access to technology being significant factors. The paper highlights the importance of computer literacy in the digital age and its varying effects across different economic regions of India.

Key Words:

Computer Knowledge, Employment, Income, Education

INTRODUCTION

In today's rapidly digitizing world, computers and smartphones have become an integral part of our lives, revolutionizing the way we work, communicate and access information. The pervasiveness of computer technology has undoubtedly reshaped economies and industries worldwide, and India is no exception. In the light of this scenario, how exactly the technical know-how impacts the employment landscape in India? What factors could potentially improve or are holding back from gaining computer skills? Are these effects uniform across rich and poor states? As we dive into these questions, we might find the dynamics between technology, employment, and income.

This paper aims at three key objectives:

- To conduct an analysis of the influence of computer skills on employment levels in India
- To explore the factors that determine the computer knowledge of an individual
- To examine the difference in outcomes of computer knowledge in high-income and low-income states

From the analysis, it is expected that:

- The level of computer knowledge possessed by an individual has a positive correlation with their probability of employment
- Factors such as education level, income, access to internet & broadband affect the computer knowledge of an individual
- There is no significant difference in impact of computer knowledge on employment in high-income and low-income states

LITERATURE REVIEW

The study by **Babina et al. (2023)** explores the impact of artificial intelligence (AI) on employment levels. It suggests that AI investments do not necessarily lead to labor-replacing effects, but rather product innovation. AI can automate tasks and create new job opportunities in AI-related fields, contributing to economic growth. The adoption of AI technologies can increase sales and employment within industries, suggesting a positive impact on job creation and economic growth. Policymakers and businesses should invest in upskilling and reskilling programs to ensure a smooth transition to AI-driven economies.

The paper "The Future of Employment" by **Frey and Osborne** discusses the impact of computerization on employment, revealing that 47% of US employment is at risk, with certain jobs being more susceptible. The study also highlights a negative relationship between computerization probability and wages and educational attainment. It also predicts a shift towards high-income cognitive jobs and low-income manual occupations, with a reduction in middle-income routine jobs.

Deming and Noray (2018) finds that STEM jobs are experiencing rapid skill obsolescence due to the rapid adoption of new technologies. The economic return on applied STEM degrees decreases within

the first decade of employment. Career shifts are notable among STEM graduates due to the need for constantly updating skills. The rapid adoption of new technologies also influences the labor market and the demand for up-to-date skills. While STEM careers start with high earnings, continuous learning and adaptation to new technologies can affect long-term career stability and wage growth.

Fairlie (2012) explores if home computer access disparities hinder the acquisition of computer skills. A randomized field experiment showed that students receiving free computers showed significantly higher levels of computer skills compared to those without. The greatest improvements were observed among young, minority, low-income, and female students. Low-income students benefited more from home computer access, likely due to fewer alternative access points and less prior exposure. Younger students under 25 may benefit more from free computers in developing computer skills.

The study by **Seyal et.al. (1999)** examines computer attitudes among non-computing academics in Brunei Darussalam's technical colleges. It found that personal computer ownership and skill level were significant factors in attitudes. A validated 12-item scale was used to measure these attitudes. The overall attitude was favourable, suggesting readiness to integrate computer technology into teaching and administrative tasks.

The study by **Volman (1997)** on gender-related effects of computer and information literacy education revealed several key findings. Changes in girls' and boys' attitudes towards computers, their knowledge of information and computer literacy (ICL), and their future plans were influenced by the curriculum materials and teaching style used in these classes. When students first entered the classroom, gender-related patterns of behaviour and attitudes were evident, but education was a major factor in forming these attitudes. The course was successful in reducing the knowledge gaps between boys and girls, but it was unable to completely eradicate the attitudes that differed based on gender. For students who worked with a non-gender-inclusive method, there was an increase in gender differences in attitudes. These results emphasise the significance of instructional strategies and curriculum design in addressing gender-related effects in computer and information literacy.

DATA HANDLING

SOURCE

This data is taken from Multiple Indicator Survey (MIS), NSS 78th Round, using Blocks 3, 4 and 5. This data is available on MoSPI website. This survey was conducted between January 2020 and December 2020.

LOADING AND MERGING DATA

1. Loading Data:

- Data from different levels (Level 2, Level 3, Level 4, Level 7) is loaded using the `read_fwf` function from the `readr` library.
- For extracting all the columns in Level 7, for the first 32 columns, the information about position of columns is taken from Level 1 in the layout file to extract columns like FSU Serial No. (Item 1), Second Stage Stratum No. (Item 5), and Sample Household No. (Item 6).

- From other levels, only the specific columns are extracted and the other columns are preserved as text data as variables named like 'other_var' or 'v1', etc.

For instance, columns extracted from Level 2 data include:

- *fsu_slno*: Column extracted from positions 1 to 5.
- *other_var_l2*: Column extracted from positions 6 to 29.
- *second_stg_stratum*: Column extracted from position 30.
- *sample_hh_no*: Column extracted from positions 31 to 32.
- *person_sl_no*: Column extracted from position 33.
- *v1*: Column extracted from position 35.
- *gender*: Column extracted from position 36.
- *other_var_l2*: Column extracted from positions 37 to 54.
- *ns_count*: Column extracted from positions 55 to 57.
- *multiplier*: Column extracted from positions 58 to 67.

Here, the only needed variables were FSU Serial No., Second Stage Stratum No., Sample Household No., Person Serial No., and Gender. Other variables were not needed, hence kept in other variables columns to save the space, and improve efficiency.

2. Creating Primary Keys:

- 2 different primary keys are created to facilitate merging of different levels of data.
- The first primary key is created by concatenating specific columns like FSU Serial No., Second Stage Stratum No., and Sample Household No.
- The second primary key is created by concatenating specific columns like FSU Serial No., Second Stage Stratum No., Sample Household No., and Person Serial No.

3. Merging Data:

- After loading and creating primary keys, the data from different levels is merged using the merge function in R.
- First, Level 7 and Level 3 are merged using the first primary key, then this is further merged with Level 2 data with second primary key and further again merged with Level 4 data with the first primary key. This forms the merged data with data from Levels 2,3,4 and 7.
- Duplicated columns are removed after merging to ensure data integrity.

FILTERING DATA

- Data is filtered based on age and enrolment status criteria, where records with age greater than or equal to 15, who were ever enrolled but not currently enrolled (i.e. Enrolment Status == 2) are selected for further analysis. Here:
 - *age*: This variable is referred from item no. 2 of Block 5 (Level 7), and
 - *enrolment_status*: This variable is referred from item no. 4 of Block 5 (Level 7)

- Further, after creating *state_code* variable (described under variable creation), based on State-wise Average Monthly Expenditure, top 5 richest and top 5 poorest states are being retained to create two more datasets respectively.

VARIABLE DESCRIPTION AND CREATION

1. Y_i (Employment Status):

This binary variable represents the employment status of an individual (i).

This variable is referred from item no. 3 in Block 5 (Level 7 data). Here, all the observations with code 2 are converted to 0.

A value of 1 indicates that the individual is employed, while a value of 0 signifies unemployment.

2. C_i (Computer Knowledge Score):

C_i is a count variable measuring the computer knowledge score of an individual (i).

This variable is referred from items 8 to 16 of Block 5 (Level 7 data), indicating various parameters of computer knowledge. These are binary variables, where all the observations with Code 2 are converted to 0 for all the 9 parameters.

A value of 1 indicates that the individual has the knowledge of that skill, while a value of 0 means otherwise, for all the 9 skills.

To create this variable, these 9 columns are summed up for each individual (i).

It ranges from 0 to 9, with higher scores indicating greater computer literacy.

3. Education Level:

It is a categorical variable representing the education level of an individual (i).

This variable is referred from item no. 6.1 in Block 5 (Level 7 data).

It is coded as follows:

3: Postgraduate and above (Code 15)

2: Undergraduate (Code 13 or 14)

1: High school (Code 10)

0: Other education levels

This variable is directly used for graphical analysis.

Further, three more dummy variables are made from this variable:

- i. *pg_and_above*: This is a dummy variable, where it is equal to 1 when $Ed_i=3$, and 0 otherwise
- ii. *ug*: This is a dummy variable, where it is equal to 1 when $Ed_i=2$, and 0 otherwise
- iii. *higher_educ*: This is a dummy variable, where it is equal to 1 when $Ed_i=1$, and 0 otherwise

4. ME_i (Household Monthly Expenditure):

ME_i denotes the total monthly expenditure of the household to which individual (i) belongs.

This continuous variable can vary widely depending on income, family size, and living standards.

It is referred from item no. 10 in Block 4 (Level 3 data).

5. G_i (Gender):

A binary gender indicator where a value of 1 represents male, and a value of 0 represents female for each individual (i).

It is referred from Block 3 (Column 4, Level 2 data). Here, all the observations with code 2 are converted to 0, and all the observations with Code 3 are removed.

6. A_i (Age):

It signifies the age of an individual (i), measured in years.

It is referred from item no. 2 in Block 5 (Level 7 data).

The merged data contains only those individuals whose age is 15 or above.

7. S_i (Social Group):

Indicates whether an individual (i) belongs to a Scheduled Caste (SC), Scheduled Tribe (ST), or Other Backward Class (OBC) social group.

It is referred from item no. 3 in Block 4 (Level 3 data). This variable is coded as follows:

$S_i = 1$, for Codes 1,2, or 3

$S_i = 0$, for Code 9

If the person belongs to SC, ST or OBC, it takes a value of 1; otherwise, it is coded as 0.

8. M_i (Access to Mass Media):

A binary indicator denoting whether an individual has access to mass media such as internet, newspaper, or magazine.

It is referred from item no. 11 Block 4 (Level 4 data). Here, all the observations with code 2 are converted to 0.

So, this variable takes a value of 1 if yes and 0 otherwise.

9. B_i (Broadband Availability):

Indicates whether the household has broadband.

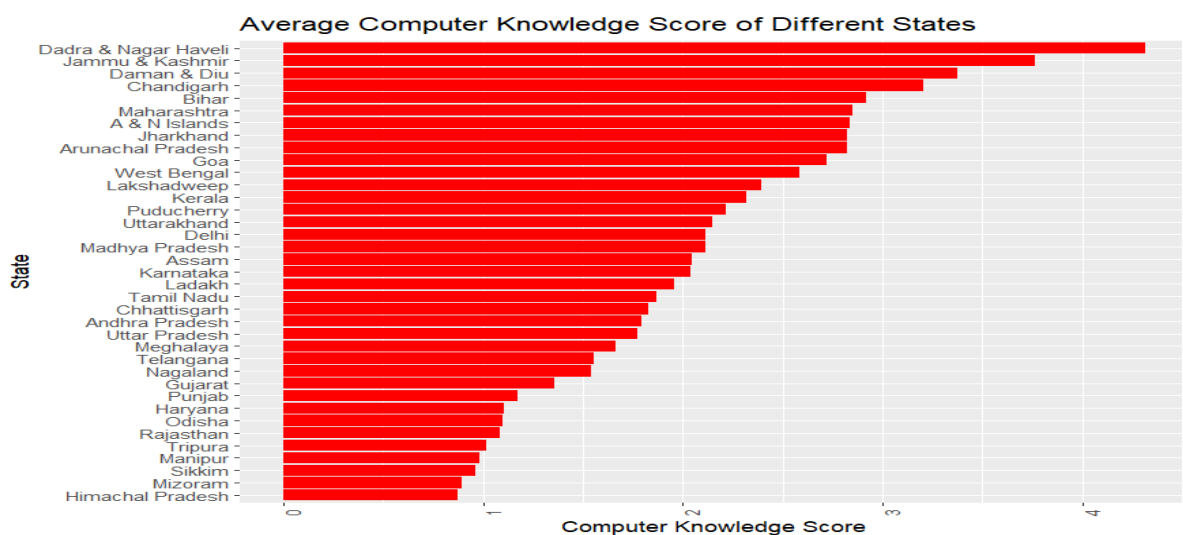
It is referred from item no. 15 Block 4 (Level 4 data). Here, all the observations with code 2 are converted to 0.

So, a value of 1 means yes, and 0 means no.

TRENDS AND PATTERNS

The average computer knowledge scores for each state are shown Figure 1. All the Indian States and Union Territories are listed on the y-axis. The computer knowledge scores are shown on the x-axis. A corresponding red horizontal bar shows the score for each state. Higher scores are indicated by longer bars. With its bar extending furthest along the x-axis, Dadra & Nagar Haveli has the highest score. Out of all the states, Himachal Pradesh has the lowest score.

FIGURE 1: STATE-WISE AVERAGE COMPUTER KNOWLEDGE SCORES



The proportion of individuals employed in each state are shown in Figure 2. All the Indian States and Union Territories are listed on the y-axis. The employment level shown on the x-axis. A corresponding green horizontal bar shows the proportion of individuals employed for each state. Higher proportions are indicated by longer bars. With its bar extending furthest along the x-axis, Tripura has the highest employment level. Out of all the states, Chandigarh has the lowest employment.

FIGURE 2: STATE-WISE PROPORTION OF INDIVIDUALS EMPLOYED

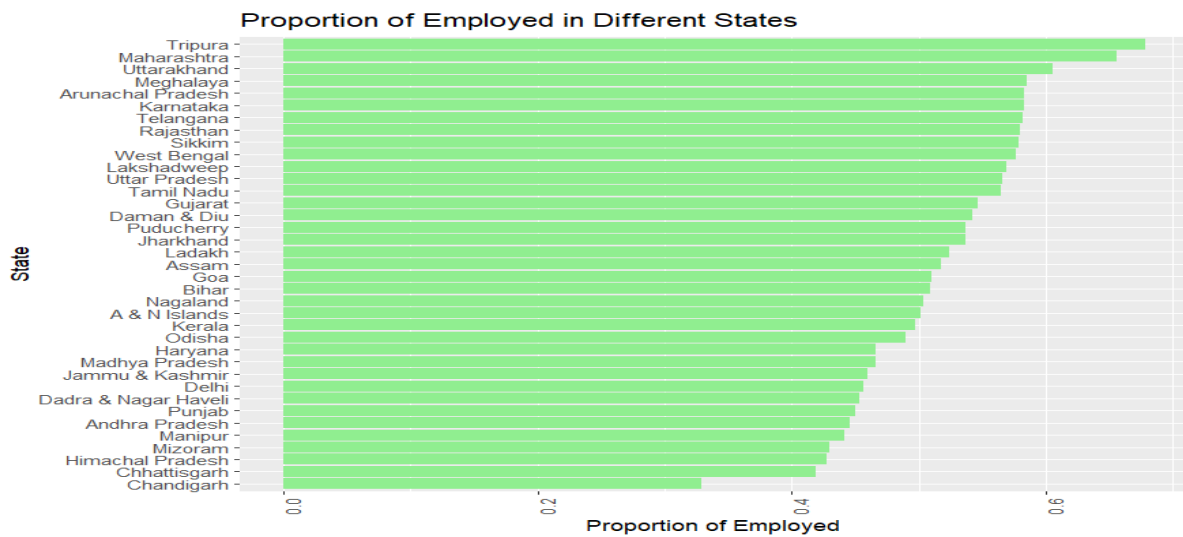


Figure 3 displays the average proportion of people in various states who have attained different levels of education: High School, Undergraduates and Postgraduate and above,

The x-axis represents the states, while y-axis represents proportion ranging from 0 to 1.

- Blue bars represent the proportion of people in a state who were enrolled in high school
- Green bars represent the proportion of people in a state who were enrolled in Undergraduate programs
- Yellow bars represent the proportion of people in a state who were enrolled in Postgraduate programs or above

States like Chandigarh, Arunachal Pradesh, Delhi-NCR, Goa, Jammu and Kashmir among others have higher proportion of people having reached to these education levels. The proportion of people enrolled in Postgraduate programs or above is very low across all the states.

FIGURE 3

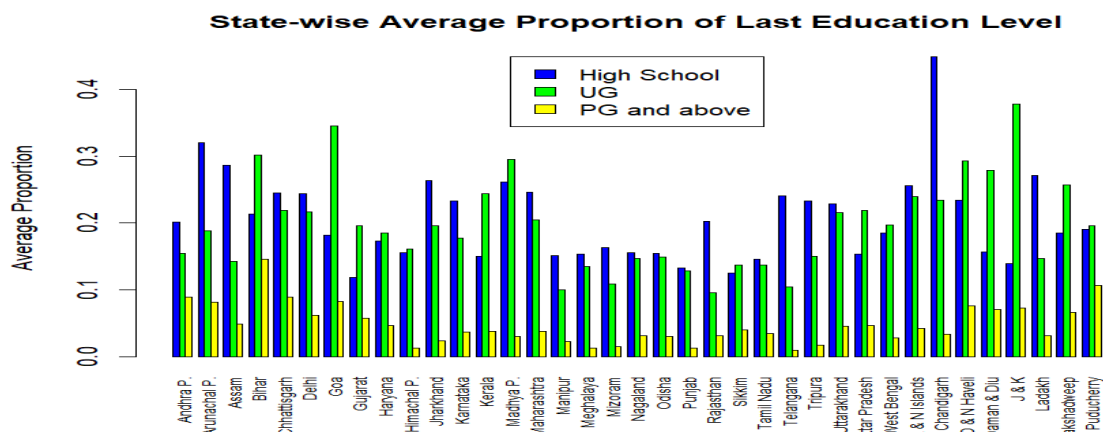
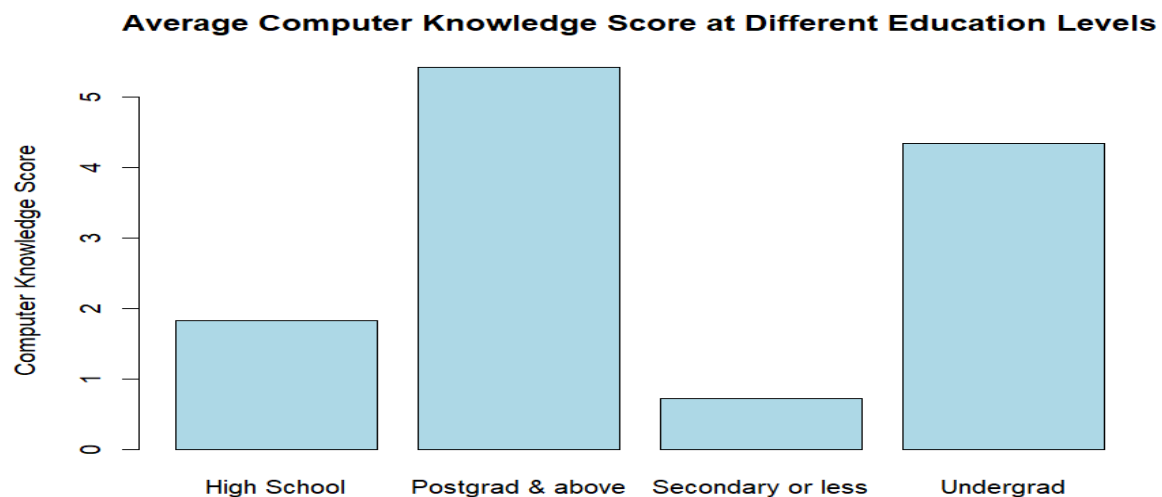


Figure 4 shows the Computer Knowledge Score across different education levels. As expected, those enrolled in Postgraduate programs and above have the highest scores with an average of slightly above 5 out of 9, followed by those in Undergraduate programs with score slightly above 4, further followed by those enrolled in high school. Those who were last enrolled in secondary education or below had the least computer knowledge on average.

FIGURE 4

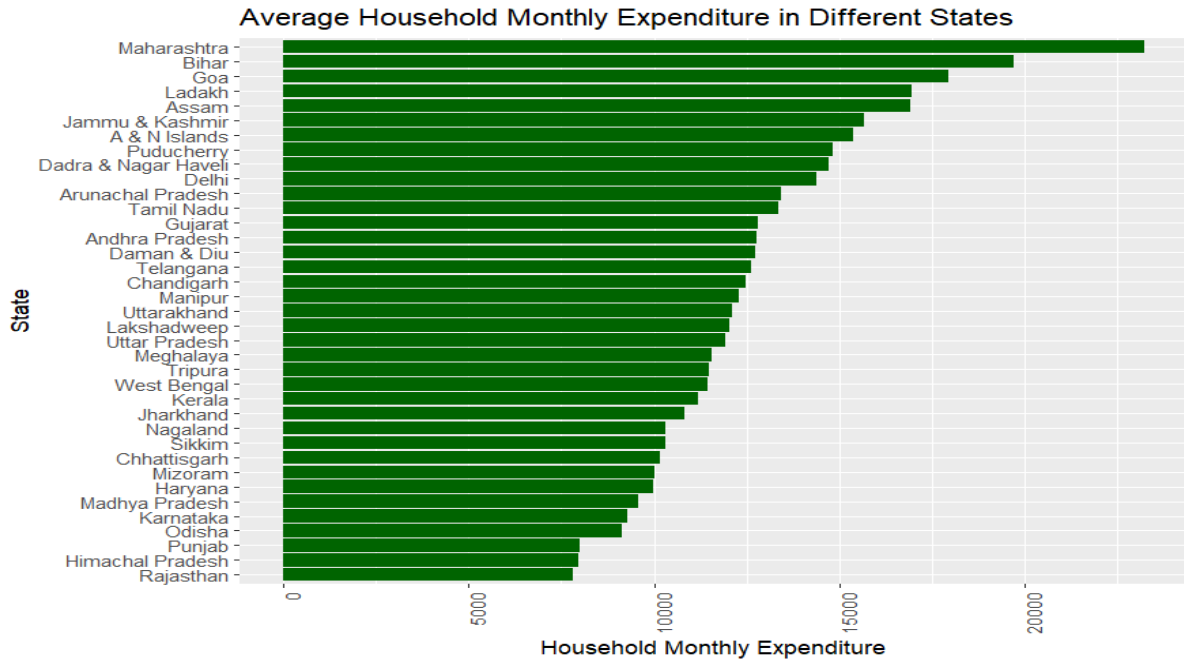


The average household monthly expenditure for each state is shown in Figure 5. All the Indian States and Union Territories are listed on the y-axis. The household monthly expenditure are shown on the x-axis. A corresponding dark green horizontal bar shows the score for each state. Higher scores are indicated by longer bars.

Maharashtra, Bihar, Goa, Ladakh and Assam are the richest 5 States/UTs in India, according to this graph as these states have the highest monthly expenditure on average. Whereas, Rajasthan, Himachal Pradesh, Punjab, Odisha and Karnataka are the poorest 5 States/UTs in India.

This classification will form the base of two categories of regression models for the Richest 5 and the Poorest 5.

FIGURE 5



METHODOLOGY AND MODELLING

1. Logistic Regression Model (First Hypothesis):

To test the 1st hypothesis, the following logistic regression model is employed:

$$Y_{1i} = \alpha_{1i} + \beta_{1i}C_i + \gamma_{1i}^{pg} Ed_i^{pg} + \gamma_{1i}^{ug} Ed_i^{ug} + \gamma_{1i}^{hs} Ed_i^{hs} + \delta_{1i}ME_i + \eta_{1i}A_i + \phi_{1i}G_i + \theta_{1i}S_i + \epsilon_{1i}$$

Dependent Variable:

Y_{1i} represents a binary outcome for individual (i)

Independent Variables:

C_i : Computer knowledge score of individual (i)

Ed_i^j : Education level (j) of individual (i), where j = pg (*pg_and_above*), ug (*ug*) and hs (*high_school*)

ME_i : Household monthly expenditure of individual (i)'s household

A_i : Age of individual (i)

G_i : Gender of individual (i)

S_i : Social group of individual (i)

β_{1i} coefficient is expected to be positive, suggesting that higher computer knowledge increases the odds of employment.

State-Specific Analysis:

The model is run separately for the 5 richest and 5 poorest states to explore regional variations.

2. Poisson Regression Model (Second Hypothesis):

To test the 2nd hypothesis, following Poisson regression model is being used:

$$C_i = \alpha_{2i} + \gamma_{2i}^{pg} Ed_i^{pg} + \gamma_{2i}^{ug} Ed_i^{ug} + \gamma_{2i}^{hs} Ed_i^{hs} + \delta_{2i} ME_i + \eta_{2i} A_i + \phi_{2i} G_i + \theta_{2i} S_i + \Phi_{2i} M_i + \psi_{2i} B_i + \epsilon_{2i}$$

Dependent Variable:

C_i represents a count variable (varying from 0 to 9) for individual (i).

Independent Variables:

All the independent variables ($Ed_i^j, ME_i, A_i, G_i, S_i$) are same as in the logistic regression model, except for the addition of M_i and B_i .

M_i : Access to mass media (internet, newspaper, etc.).

B_i : Broadband availability in the household.

State-Specific Analysis:

Similar to the logistic regression, this model is applied separately for the 5 richest and 5 poorest states.

RESULTS

Table 1 shows the effect of computer knowledge on likelihood of employment of an individual. It shows that overall, for each unit increase in score, the odds of being employed increase by 5.7%, and this coefficient is significant at 1% level. This highlights the importance of computer knowledge. This coefficient varies between the richest 5 and the poorest 5 states. This is to be confirmed by t-test, whether this difference is significant or not.

However, the coefficients of education (PG, UG and High School) are surprisingly negative and significant. Possible explanations could be poor labour market conditions in 2020, during which this survey took place or simply a sampling bias.

The remaining result pattern remains consistent among all the groups.

TABLE 1: EFFECT OF COMPUTER KNOWLEDGE ON EMPLOYMENT STATUS

	<i>Dependent variable:</i>		
	employment_status		
	Overall	Richest 5	Poorest 5
comp_know_count	0.057*** (0.003)	0.102*** (0.010)	0.032*** (0.008)
higher_educ	-0.463*** (0.014)	-0.369*** (0.051)	-0.449*** (0.038)
ug	-0.992*** (0.017)	-0.860*** (0.066)	-1.050*** (0.046)
pg_and_above	-0.641*** (0.029)	-0.383*** (0.102)	-0.631*** (0.103)
hh_monthly_exp	-0.00001*** (0.00000)	0.00000 (0.00000)	-0.00003*** (0.00000)
gender1	3.090*** (0.011)	2.527*** (0.042)	3.283*** (0.031)
age	0.133*** (0.001)	0.079*** (0.004)	0.156*** (0.003)
soc_grp_var	0.228*** (0.012)	0.420*** (0.043)	0.399*** (0.037)
Constant	-4.973*** (0.034)	-3.454*** (0.132)	-5.873*** (0.092)
Observations	268,687	15,094	40,473
Log Likelihood	-122,124.500	-7,716.372	-17,663.890
Akaike Inf. Crit.	244,266.900	15,450.750	35,345.780

Residual Deviance 244,248.900 (df = 268678) 15,432.750 (df = 15085) 35,327.780 (df = 40464)

Null Deviance 372,063.700 (df = 268686) 20,742.680 (df = 15093) 56,034.290 (df = 40472)

Note: *p<0.1; **p<0.05; ***p<0.01

Table 2 shows how various determinants affect the computer knowledge of an individual. Higher education level is associated with significantly higher computer knowledge. Those who have education of PG and above show the highest odds. Monthly household expenditure also shows the positive impact on computer knowledge. Availability of mass media items like internet, newspapers, magazines, etc. and broadband services in a household also positive impact on computer knowledge of an individual.

Similar patterns are shown across the richest and the poorest states with a slight difference in magnitude.

TABLE 2: DETERMINANTS OF COMPUTER KNOWLEDGE

	<i>Dependent variable:</i>		
	comp_know_count		
	Overall	Richest 5	Poorest 5
higher_educ	0.493*** (0.005)	0.480*** (0.016)	0.610*** (0.014)
ug	1.011*** (0.004)	0.998*** (0.015)	1.003*** (0.012)
pg_and_above	1.174*** (0.005)	1.173*** (0.020)	1.279*** (0.019)
hh_monthly_exp	0.00001*** (0.00000)	0.00001*** (0.00000)	0.00000*** (0.00000)
soc_grp_var	-0.005 (0.003)	0.057*** (0.011)	-0.072*** (0.011)
age	-0.008***	-0.011***	-0.011***

	(0.0003)	(0.001)	(0.001)
gender1	0.357***	0.269***	0.517***
	(0.003)	(0.011)	(0.010)
mass_media	0.419***	-0.048	0.192***
	(0.010)	(0.043)	(0.019)
broadband	2.359***	1.988***	2.401***
	(0.007)	(0.026)	(0.018)
Constant	-2.246***	-1.270***	-2.148***
	(0.014)	(0.056)	(0.035)
Observations	268,687	15,094	40,473
Log Likelihood	-375,866.000	-25,921.930	-43,872.420
Akaike Inf. Crit.	751,752.000	51,863.860	87,764.850
Residual Deviance	400,259.100 (df = 268677)	25,065.260 (df = 15084)	52,512.360 (df = 40463)
Null Deviance	983,163.000 (df = 268686)	53,087.570 (df = 15093)	126,349.300 (df = 40472)
Note:	*p<0.1; **p<0.05; ***p<0.01		

T-TEST

To check if there is significant difference between the computer knowledge coefficients (Table 1) of top 5 richest and 5 poorest states in India, a two-tailed two-sample t-test is being used here

The null hypothesis is that the coefficient of Computer Knowledge Score is not statistically different in the 5 richest and in 5 poorest states

t-statistic = 5.597 and p-value is nearly 0

There is sufficient evidence to reject the null hypothesis at 1% significance level

The effect of Computer Knowledge on employment is much higher in richest states than in the poorest states (t-statistic > 0)

CONCLUSION

From this paper, it can be concluded that employment levels vary with the level of computer knowledge each individual has. Various factors such as education level, income (or household consumption expenditure), availability of mass media or broadband or other technologies affect the computer knowledge of an individual. It is also observed that the impact of computer knowledge varies across high-income and low-income states.

This paper can serve as a model to evaluate the effect of different programs affecting availability of technology at home or computer awareness programs. This can also be extended to studies in various technical or non-technical fields.

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